

OCOsense Glasses for Facial Expressions Recognition and Contextual Affective Computing in Real World and Augmented Reality

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ABSTRACT

The paper presents the novel OCOsense™ smart glasses with integrated sensors, primarily non-contact optomyographic (OMG) OCO™ sensors, 9-axis inertial measurement unit (IMU), and an altimeter. The glasses connect with a smartphone application, which facilitates the continuous and real-time measurements of facial-muscles activation and head movement, thus allowing for the detection of facial expressions and the activities of the user in real-time. We will demonstrate how the system is used in practice, i.e., a participant will wear the OCOsense™ glasses, which will stream the sensor data to a tablet, where the real-time visualization of the sensor data and the data interpretation will be presented such as facial expressions (smile, frown, surprise) and activities. We believe that the OCOsense™ glasses are the next big thing in wearables, which will allow for better understanding of the user's context, activities, emotional state, and more, which can be easily coupled within Augmented and Extended Reality environments.

CCS CONCEPTS

• Machine Learning; • Sensors and actuators; • Ubiquitous and mobile computing;

KEYWORDS

Glasses, Affective Computing, Facial Expressions, Emotion Recognition, Activity, Valence, OMG, IMU, Machine Learning

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1 FACIAL EXPRESSION RECOGNITION

The face is one of the most expressive mediums of our body[1], communicating emotional and mental states, as well as behavioural intentions, via facial muscle activations and configurations which in turn construct expressions. The behavioural research on facial expressions has been an active topic from Darwin 1872 [2], and Duchenne 1876 [3] to today, featured in neuro-psychological, and detection related studies using camera- and sensor-based systems, such as electromyography (EMG).

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The field of emotion recognition from facial configurations and activations is steadily progressing, utilizing advances on the sensor processors, miniaturization, wearability, and connectivity. In terms of detection accuracy, EMG sensors were primarily used to detect muscle activations on the face [4], and it is considered as the highest resolution and ‘gold-standard’ method, upon which newer techniques are validated. Camera-based systems are commonly used for facial tracking and for inferring emotional states in the wild, albeit its efficacy has been greatly debated [5] and often privacy concerns were expressed when such approaches were utilised in the public. Due to limitations related to sensor placement, lighting, intrusiveness, and movement, measuring emotional expressions naturalistically in the wild remains a challenge.

Although wearable sensing systems exist for the recording of physiological markers, for example in the form of smartwatches, there are no methods currently available to detect and discriminate facial expressions and valence (see definition in [6]) in the wild without a camera or sensors placed directly on the skin. As such, existing sensing methods are obtrusive, conspicuous by necessity and difficult to engage with in remote and free-field settings on a daily basis.

To address these challenges, we developed a novel, multimodal sensor-enabled glasses - the OCOsense™ (shown in Figure 1). OCOsense™ is a pair of wearable glasses with integrated OCO™ sensors based upon non-contact optomyographic (OMG) technology [7], smart 9-axis inertial measurement unit (IMU) including an altimeter. The glasses sensors developed can be paired with an application for smartphones, which facilitates the continuous and temporally dense (i.e., second-by-second) measurement of facial muscle displacement and head motion, and in turn detects facial expressions and activities in real-time. We will demonstrate how the system is used in practice, the steps required to set-up and the demo environment within the application we have built for recording facial and head-based responses to video stimuli. Such wearable technology could substantially simplify emotional state recognition in the wild, within Augmented Reality (AR) and real-world environments.

A key innovation in OCOsense™ is the potential to address known limitations in real-world emotion detection by linking a facial expression to the context in which it is performed [8] (see Figure 2, right panel). Our prototype demo application allows the recording of data within a stimuli-presentation contained screen, similar to well-known mood induction study paradigms, and replays the experience back along with the sensor data and expressions detections using our proprietary Affective Artificial Intelligence (AI) Engine, in a moment-by-moment fashion.

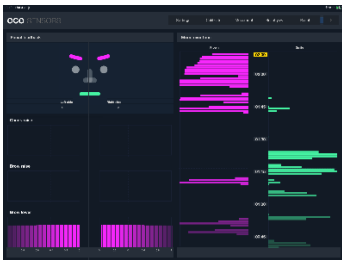


Figure 1: The OCOsense™ Glasses with detachable outward facing camera.

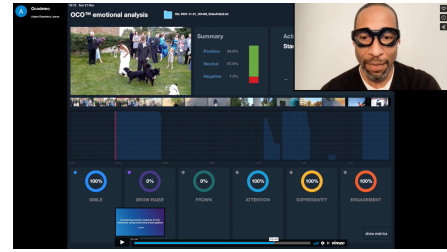
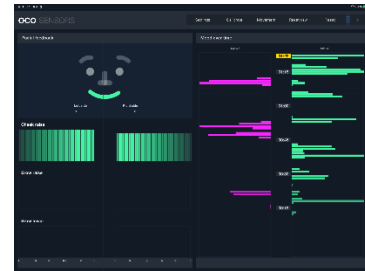


Figure 2: The OCOsense™ mobile application, showing (i) the expression detection interface when a user is frowning and smiling (left) and the stimuli emotional analysis screen replaying a pre-recorded experience (right).



2 OCOSENSE™ SYSTEM

The OCOsense™ system consists of a: (i) multi-sensor wearable glasses (shown in Figure 2); and the (ii) mobile application; which are explained below.

The OCOsense™ glasses (shown in Figure 1 and Figure 3) consist of a novel smart multi-sensor wearable device that can detect emotional state, expression, and facial muscle activation through its integrated non-contact sensors inside the frame of the glasses. The OCO™ sensors are an optical non-contact OMG approach that can read facial movement in 3-axis providing higher resolution signal ($\pm 4.7\mu\text{m}$ X&Y-axis, $\pm 4.0\mu\text{m}$ Z-axis) than average camera-based solution and has advantages over EMG-based systems. EMG electrodes require firm and constant contact with skin in order to achieve an acceptable signal to noise ratio, which is not possible in a glasses format; instead, the OCO™ sensors are optically based, therefore they don't require skin contact and can function accurately from 4mm to 30mm away from the skin. OMG has far superior signal to noise ratio compared to EMG and does not require a complex

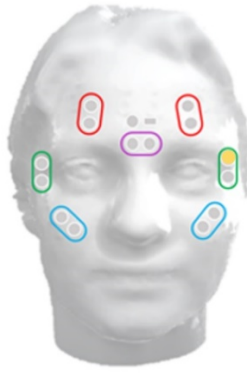


Figure 3: OCO Sense™ glasses on an average head model (left) and the sensor locations shown upon a 3D rendered face (right). The coloured rectangles represent the OCO™ sensors within the frame, upon major facial muscles of the face, namely frontalis (in red), zygomaticus (in blue) and corrugator (in purple). The green OCO™ sensors positioned over the temples are used to determine movement artifacts in relation to the head. In addition, there is a 9-axis smart IMU and altimeter that are located in the right arm of the glasses frame indicated by the yellow circle.

filtering procedure to extract a useable signal which is computationally expensive, rendering it as a strong candidate for wearable integration. Additionally, the OCO™ sensors are designed to provide output data for three 16-bit coordinates positions (X, Y and Z) per sensor and stream data at 50Hz for this application, resulting in a data rate per sensor of 2.4kb/s. This is a tenfold reduction in data rate compared to a standard 24-bit single differential channel facial EMG based system with minimum sample rate of 1000Hz (i.e. 24kb/s).



The OCO Sense™ application (see Figure 2 and Figure 4) is a software currently developed for iPad iOS and Mac OS. The app connects via Bluetooth LE to the glasses. After successful pairing has been achieved, the app receives and visualises via graphs all sensor data from the device in real-time. emteq’s Affective AI Engine first uses calibration techniques to analyse the person’s face, and then provides high-level interpretations including expression types, intensity and activities performed. As part of the calibration, the user performs an expression such as “smile” 3 times, lasting overall <1 minute. The demo screens developed visualises activity and expressions recognition in real-time, using a digital avatar face and amplitude graphing of the OCO™ sensor via the locally running the Affective AI Engine.

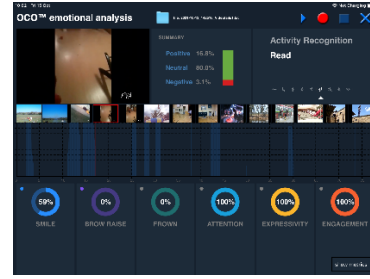
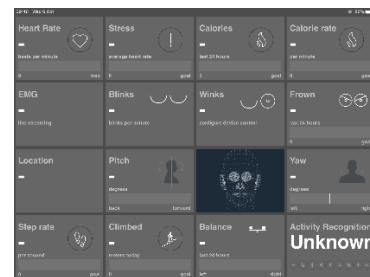


Figure 4: Main screen showing expressions and activity recognition screen (left) and the demo experience analysis screen (right).

3 DEMO

The demo will show how the system is used in practice. For these purposes, we have developed a short stimuli-experience specifically designed to induce various emotional responses to users (Figure 4). First, we will show how the OCO Sense™ glasses are placed on the face and how to check the glasses’ fit. Then, we will show the application we developed, where the user can choose different options and calibrate their face to the sensors. Next, we will show how the data is recorded and visualized in the app in real-time. Finally, we will show the output of the expression detection Affective AI Engine that provides the insights about the user’s affective state, i.e., valence and expressions along with the contextual information for the short stimuli experience.

The system can have various use cases, starting from researchers interested in affective computing and expression detection, within real-world scenarios or longitudinal studies where comfort and wearability is very important. The data output could also be using in the adaptive control of applications, for example gesture-based human-computer interaction. Lastly, our team envision the glasses being used on a daily basis, to monitor and provided feedback to users in various settings for health-care and well-being.



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REFERENCES

- [1] P. Ekman and W. V. Friesen, *Facial Action Coding System: A Technique for the Measurement of Facial Movement*. Palo Alto: Consulting Psychologists Press, 1978.

- [2] [2]C. Darwin, *The expression of the emotions in man and animals (3rd ed.)*. 1872.
- [3] [3]G. Duchenne, *Mécanisme de la physionomie humaine: où, Analyse électro-physiologique de l'expression des passions*. 1876.
- [4] [4]A. J. Fridlund and J. T. Cacioppo, "Guidelines for Human Electromyographic Research," *Psychophysiology*, vol. 23, no. 5, pp. 567–589, 1986, doi: 10.1111/j.1469-8986.1986.tb00676.x.
- [5] [5]L. F. Barrett, R. Adolphs, S. Marsella, A. M. Martinez, and S. D. Pollak, "Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements," *Psychol. Sci. Public Interes.*, vol. 20, no. 1, pp. 1–68, Jul. 2019, doi: 10.1177/1529100619832930.
- [6] [6]J. Russell and L. Barrett, "Core affect, prototypical emotional episodes, and other things called emotion: dissecting the elephant," *J. Personal. Soc. . . .*, 1999, [Online]. Available: <http://psycnet.apa.org/journals/psp/76/5/805/>.
- [7] [7]H. H. Muhammed and J. Raghavendra, "Optomyography (OMG): A Novel Technique for the Detection of Muscle Surface Displacement Using Photoelectric Sensors," *diva-portal.org*, vol. 10, no. 13, 2015, Accessed: Aug. 03, 2022. [Online]. Available: [https://www.diva-portal.org/smash/record.jsf?pid\\$=diva2:886303](https://www.diva-portal.org/smash/record.jsf?pid$=diva2:886303).
- [8] [8]Lisa Feldman Barrett, *How Emotions Are Made by Lisa Feldman Barrett*. 2018.